

Kaveri Sharma

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EDUCATION

PES University

Bachelor of Technology in Computer Science

– Relevant coursework: Data Science, ML & Database Systems, Data Structures & Algorithms, Web Development

Bengaluru

Expected 2027

EXPERIENCE

Data & Analytics Engineering Intern

Jun 2025 – Aug 2025

National Capital Region Transport Corporation (NCRTC)

New Delhi (On-site)

- Built and deployed a real-time transport analytics dashboard for the PM e-Bus Sewa initiative, unifying 3+ heterogeneous fleet/ridership sources into a single CMS-integrated UI for NCR stakeholders
- Translated government policy KPIs into interactive visualization workflows, cutting stakeholder time-to-insight from hours to minutes and removing analyst dependency for daily reviews
- **Tech:** Data Visualization, Web Development, CMS Integration

PROJECTS

OptiSolve | *React, FastAPI, LangChain, ChromaDB, Groq/SambaNova*

github.com/.../Optisolve

- Developed a 3-tier customer support orchestration platform (React, FastAPI, LangChain) routing tickets across auto-resolve, AI-assisted, and escalate tiers based on AI confidence scores and real-time sentiment analysis
- Designed a RAG pipeline (ChromaDB, Groq Llama 3.3 for drafts, SambaNova/Llama 3.1 for sentiment scoring) with a human-in-the-loop engine updating the knowledge base from agent corrections — achieving a 30–40% reduction in MTTR
- Built a pre-submission “Gatekeeper” layer that analyzes drafts in real time and suggests fixes before ticket creation, achieving 20–30% ticket deflection

CityOS (Spatial Logistics Engine) | *Python, React, AMD ROCm, Sentinel-2, OSM*

github.com/.../Spatial-Engine

- Built CityOS, a GovTech spatial logistics engine using AMD-accelerated graph routing across 300,000+ street nodes to minimize urban inequality (Spatial Gini Coefficient), accelerating computations 4,064x (14 hrs → 12 sec) on AMD ROCm GPUs
- Replaced outdated census data by integrating Sentinel-2 satellite imagery and OpenStreetMap to map informal settlements using true walkable street distances instead of straight-line estimates
- Deployed a React-based live dashboard with 5-second GPU-accelerated re-optimization (8.5x faster than CPU), generalized beyond schools to healthcare, fire stations, and warehouse siting across any searchable city

Personalized Digital Twin for Heart Failure Prediction

- Built a heart-failure decompensation monitoring system integrating the Kitware Pulse physiology simulation engine in a Dockerized pipeline, translating patient vitals (HR, SpO2, BP, RR) into simulated trajectories via a rule-based clinical triage layer
- Designed an interpretable HF staging and risk-scoring model (LOW/MODERATE/HIGH across NYHA stages A–D), choosing a hand-tuned weighted scorer over a black-box XGBoost model, plus 7/14/30-day risk trajectory forecasting
- Engineered a synthetic data pipeline (clinical baselines from published literature/public datasets, wearable trends across stable/deteriorating/recovering regimes) and ran preprocessing, training, and evaluation on an experimental XGBoost classifier, prioritizing high recall to minimize missed high-risk cases
- Currently authoring a research paper formalizing this interpretable risk-scoring methodology for peer-reviewed publication

ACHIEVEMENTS

National 2nd Place, AMD Slingshot Challenge

May 2026

- Secured a cash prize of INR 3,00,000, ranking 2nd among 40,000+ registrations and 4,000+ competing prototypes for engineering a highly scalable geospatial AI solution — CityOS

Shortlisted, Atos SRIJAN Hackathon

May 2026

- Ranked among winning teams out of 500+ participants, earning direct internship consideration for building a customer support orchestration platform — OptiSolve

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, JavaScript, TypeScript

Data & ML: PyTorch/Keras, TensorFlow, Pandas, NumPy, Scikit-learn

Full Stack & App Integration: Next.js, React, REST APIs, JSON, Data Visualization

Tools & Databases: MySQL, Firebase Firestore, Git/GitHub, Docker

Generative AI & NLP Concepts: LLMs, RAG, NLP, Prompt Engineering, Hugging Face

LEADERSHIP & ACTIVITIES

Web Development Domain Head

Oct 2024 – Present

Encode AI

- Organized workshops on Emerging AI Tech and Rapid Development Frameworks, democratizing AI skills for 100+ students

Research and Content Domain Head

Oct 2024 – Present

ACM-W

- Researched trending topics and designed engaging challenge formats for hackathons, and curated messaging, headlines, and speeches for club events